Avinash Singh

Masters Thesis Scania/Traton

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An analytically driven statistician with a deep foundation in mathematical statistics and machine learning from Stockholm University's Department of Mathematics. Specializes in analyzing model stability, investigating underlying statistical assumptions, and implementing uncertainty quantification frameworks. Proven experience in reasoning about complex systems and evaluating model limitations to deliver robust, interpretable data solutions.

Skills

Statistical Foundations:	Probability Theory, Statistical Inference, Uncertainty Quantification, Conformal Prediction, Hypothesis Testing, Statistical Analysis
Machine Learning:	Deep Learning, Reinforcement Learning (PPO, DQN), Transfer Learning, Latent Space Analysis, Predictive Modeling
Software & Ecosystem:	Python, SQL, PyTorch, TensorFlow, Pandas, NumPy, Apache Spark, Linux, Git, AWS

Professional Experience

- **Traton / Scania Group** — Södertälje, Sweden
  Master's Thesis Worker – Autonomous Driving Post-Perception & Planning Research Team — 01/2026 – 06/2026
  - Evaluated the robustness and structural limitations of the UniAD foundational framework when subjected to significant domain shift from passenger cars to heavy-duty vehicles.
  - Applied Conformalized Quantile Regression (CQR) to develop rigorous predictive uncertainty intervals, introducing verifiable statistical safety guarantees to trajectory planning.
  - Investigated model performance disparities to demonstrate the mathematical efficiency and stability of Transfer Learning over training models from scratch.
  - Utilized Gaussian Splatting techniques to synthesize novel environmental trajectories, expanding data coverage for extreme edge-case validation.
- **Alyst AB** — Stockholm, Sweden
  Summer Data Science Intern — 06/2025 – 09/2025
  - Engineered an automated information extraction and multi-threaded report summarization pipeline utilizing the Gmail API and Gemini Flash LLM.
  - Analyzed textual patterns and metadata across hundreds of active conversation layers to generate condensed, structurally stable weekly summaries.
  - Monitored and refined model parsing performance to ensure consistency, data integrity, and robust text-processing behavior under sparse information constraints.

Methodological & Technical Projects

- **Comparative Analysis of Reinforcement Learning Policies for Game AI** — Independent Project
  - Conducted a thorough empirical evaluation comparing Proximal Policy Optimization (PPO) and Deep Q-Networks (DQN) within complex, simulated environments.
  - Investigated the root algorithmic causes behind convergence behaviors, mapping PPO's 40% performance advantage directly to the mathematical clipping of its policy updates.
  - Validated model stability limits to demonstrate how constrained policy updates systematically prevent the chaotic divergence typical of standard deep Q-learning methods.
- **Semantic Segmentation of Aerial Drone Imagery** — Computer Vision Project
  - Built a U-Net encoder-decoder network optimized for dense, pixel-level categorical classification of highly variable geographical terrain.
  - Evaluated model error distributions across high-contrast spatial boundaries (architectural, living, and environmental classes) to ensure categorical robustness.

EDUCATION

- **Stockholm University** Stockholm, Sweden
  M.Sc. in Mathematical Statistics and Machine Learning – Department of Mathematics 09/2024 – Present
  - Rigorous emphasis on core probability theory, statistical inference, data structures, and the mathematical properties governing deep neural architectures.
- **National Institute of Technology (NIT)** Srinagar, India
  B.Tech in Electrical Engineering 2011 – 2015
  - Completed core curriculum focused heavily on linear algebra, calculus, numerical methods, and differential modeling.